

Thermal physics · Mock midterm test · Version C · Solutions

Total 44 marks: I = 8, II = 14, III = 14, IV = 8. Marks for each part are shown on the right. A mark in square brackets [1] is one point earned by the step it follows. If you used a wrong value consistently in later steps (error carried forward), you still earn the later method marks. Section IV is answered in Italian.

I · Multiple choice

[8]

#	Answer	Marks
1	C — Temperature is a state quantity, while heat exists only as energy transferred because of a temperature difference.	1
2	D — X and Y are at the same temperature, as the Zeroth Law implies.	1
3	B — 1000 °C	1
4	A — It is the lowest temperature possible: no body can be cooled below it.	1
5	D — Using a narrower capillary tube, with the same bulb and the same liquid.	1
6	A — 6.0 mm	1
7	C — The washer fits more loosely, because its hole becomes larger.	1
8	B — 0.60 %	1

II · Paper 6 on the thermistor data

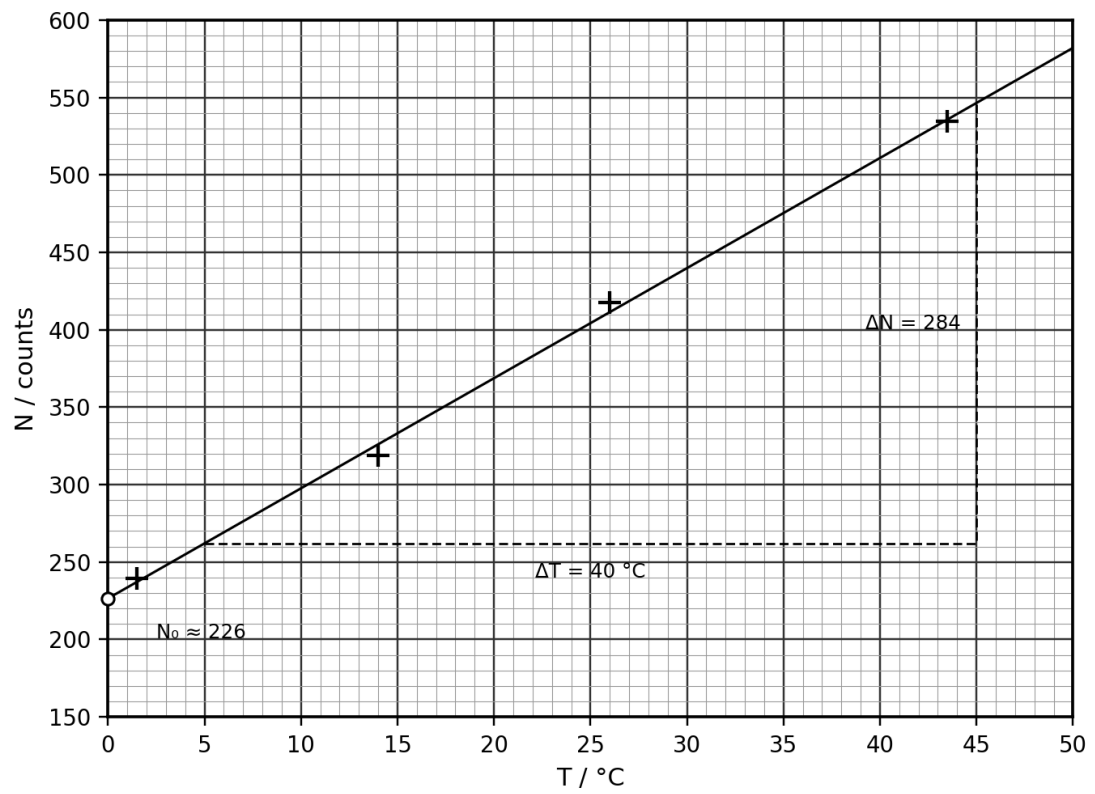
[14]

Data with the anomalous reading in square brackets and the correct means (the mean for bath 4 leaves out 494):

Bath	T / °C	N ₁	N ₂	N ₃	N ₄	N ₅	mean N
1 (ice water)	1.5	238	238	240	242	240	239.6
2 (cold water)	14.0	318	319	318	318	322	319.0
3 (room temperature)	26.0	417	420	417	417	417	417.6
4 (hot water)	43.5	536	533	534	535	[494]	534.5

Least-squares line through the four means: $m = 7.11 \text{ counts/}^\circ\text{C}$, $N_0 = 226.5$, so $T = (N - 226.5) / 7.11$. A line drawn by hand will give slightly different values (m within about $0.5 \text{ counts/}^\circ\text{C}$, N_0 within about 8 counts is fine). The test bath ($N = 490$) gives $T = 37.04 \text{ }^\circ\text{C}$. The reference thermometer reads $36.8 \text{ }^\circ\text{C}$: difference $0.24 \text{ }^\circ\text{C}$.

	Answer	Marks
(a)	Independent variable: temperature T of the bath (read on the reference thermometer). Dependent variable: thermistor reading N.	2
(b)	The reading 494 in bath 4 (N ₅), because it differs from the other four readings of that bath by about 41 counts, while those agree to within 3 counts.	1
(c)	Mean of the four remaining readings: 534.5.	1
(d)	Axes labelled "T / °C" (x) and "N / counts" (y). Four points plotted to within half a small square, each marked with a small cross (the centre of the cross is the point).	3
(e)	A single, straight, thin line drawn with a ruler, passing as close as possible to all the points, with points on both sides of it (not dot-to-dot, not forced through the first and last points).	1
(f)	Triangle drawn with a base of at least half the line (e.g. T = 5 to 45 °C); $m = \Delta N / \Delta T = 7.11 \text{ counts/}^\circ\text{C}$.	2
(g)	$N_0 \approx 227$ (read at T = 0 °C; fit: 226.5). Equation $T = (N - 226.5) / 7.11$.	1
(h)	$T = (490 - 226.5) / 7.11 \approx 37.0 \text{ }^\circ\text{C}$ (accepted: 35.5–38.5 °C).	1
(i)	Meets the requirement: $ 36.8 - 37.0 = 0.24 \text{ }^\circ\text{C} < 1 \text{ }^\circ\text{C}$.	1
(j)	One specific improvement, e.g.: take several readings in the test bath; use a reference thermometer with smaller divisions (0.1 °C resolution); stir the bath and wait for equilibrium before reading; use more calibration baths; use a second independent test bath.	1



Reference graph: points = means, least-squares line, gradient triangle from $T = 5$ to $T = 45$ °C, intercept N_0 on the axis.

III · Structured questions

[14]

	Answer	Marks
1(a)	(i) $21\text{ }^{\circ}\text{C} = 294\text{ K}$; (ii) $250\text{ K} = -23\text{ }^{\circ}\text{C}$.	2
1(b)	Heat is not something a body has. Temperature is a property (state quantity) of the body; heat is energy transferred because of a temperature difference. What the body has is internal energy. [1] heat = energy in transfer because of ΔT (a process, not stored). [1] the body has a temperature / internal energy instead.	2
1(c)	0 K (absolute zero) is the lowest possible temperature; nothing can be cooled below it, so -1 K does not exist.	1
2(a)	$0\text{ }^{\circ}\text{C}$: melting point of pure ice. $100\text{ }^{\circ}\text{C}$: boiling point of pure water (at standard atmospheric pressure).	2
2(b)	The column drops slightly at first (1): the glass bulb warms up and expands before the liquid does, so the space inside the bulb increases. Then the liquid warms up and expands more than the glass, so the column rises (1).	2
3(a)	$\Delta T = 40 - (-10) = 50\text{ }^{\circ}\text{C}$ [1]; $\Delta L = \alpha L_0 \Delta T = 12.0 \times 10^{-6} \times 8.0 \times 50$ [1] $= 4.8 \times 10^{-3}\text{ m} = 4.8\text{ mm}$ [1].	3
3(b)	$\beta = 2\alpha = 24.0 \times 10^{-6}\text{ }^{\circ}\text{C}^{-1}$ [1]; $\Delta A = \beta A_0 \Delta T = 24.0 \times 10^{-6} \times 1.20 \times 50 = 1.44 \times 10^{-3}\text{ m}^2$ (14.4 cm^2) [1].	2

IV · Problema — Anello di rame su un albero d'acciaio**[8]**

	Answer	Marks
(a)	Aumenta: il foro si dilata come se fosse fatto dello stesso materiale (come un disco pieno); tutte le lunghezze del pezzo scalano con la stessa frazione.	1
(b)	Diametro minimo del foro: $30,03 + 0,02 = 30,05$ mm [1]. $\Delta D = 0,05$ mm, quindi $\Delta T = \Delta D / (\alpha D) = 0,05 / (17,0 \times 10^{-6} \times 30,00) = 98,0$ °C [1]. $T = 20 + 98,0 = 118,0$ °C, minore del limite di 200 °C: compatibile [1].	3
(c)	Ogni lunghezza cresce di $\Delta D/D = 0,167$ %; l'area, che dipende dal quadrato del diametro, cresce circa del doppio: $\Delta A/A \approx 2 \alpha \Delta T = \beta \Delta T = 0,33$ % [1 per $\beta = 2\alpha$, 1 per il risultato]. Non serve conoscere il diametro ($\Delta A \approx 2,36$ mm ² per $D = 30,00$ mm).	2
(d)	Interferenza a 20 °C: 0,03 mm. A 150 °C: albero 30,0768 mm, foro 30,0663 mm, interferenza 0,0105 mm [1]. Resta serrato ma meno (da 0,03 a 0,0105 mm), perché il rame ha α maggiore dell'acciaio; si annullerebbe solo a 220,5 °C [1].	2

Dati utili e dati di disturbo. Servono: diametri, gioco richiesto, coefficienti α , limite del costruttore (per la verifica in (b)), temperatura di esercizio (per (d)). Non servono: massa, altezza o spessore del pezzo, potenza del forno, tempo di riscaldamento, pezzi prodotti, giri/min.